

Module 06: Learning in Middle School

Learning Objectives

1. Define **Learning** within the context of Middle School.
2. Analyze how Learning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Middle School to the formal definition of Learning.

Introduction

This module explores **Learning**. In the **Middle School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Learning is a critical component of the 8-part Active Inference spine, bridging the gap between Action and Communication.

Key Concepts

1. Learning as a Markov Blanket Boundary

How does Learning define the boundary between the agent and the environment?

2. Generative Models of Learning

What parameters involved in Learning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Learning drive the perception-action loop?

Applications

In Middle School, we see Learning manifest in: * **Specific Example 1:** Training an AI to recognize handwritten digits works just like human learning through prediction errors. The

model starts by guessing randomly, gets told "that 7 is not a 1," and adjusts its internal weights to make fewer mistakes next time. After seeing thousands of examples and correcting thousands of prediction errors, it builds a model accurate enough to read messy handwriting that even some humans struggle with. * **Specific Example 2:** When you learn a new programming language after already knowing one, your brain transfers its existing model and then updates it where the languages differ. If you know Python and start learning JavaScript, you predict that `print()` will work -- but it does not, so you learn `console.log()` instead. Each prediction error from your old model helps you build the new one faster, which is why the second programming language is always easier than the first.

Conclusion

Understanding Learning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.